

Learned miRNA→gene regulatory model — design report

Responding to `LEARNED_MODEL_DISCUSSION_PROMPT.md` (§9 deliverable). **Scope.** How to *learn* the miRNA-arm → Hallmark-gene regulatory weights from TCGA-BRCA expression under multi-source evidence priors, how they aggregate/attribute, and how to know they capture real targeting. The current softmax/D(m)/edge_w machinery is treated as **one baseline among several**, per §0.

Status. Design / not yet built. This is the reasoning-first spec to audit and act on. Companion to `FORMULAS.md` (coded heuristic), `LEARNED_REGULATORY_MATRIX_DESIGN.md` (earlier RRR sketch, which this supersedes and extends), `DATA_SOURCES.md`, `MODELING_FRAMEWORK.md`, `EDGE_QUESTION_TAXONOMY.md`, `GENE_QUESTION_TAXONOMY.md`.

Legibility convention. I mark **[assert]** where a claim is established by literature/data cited here, **[infer]** where I am reasoning beyond direct evidence, and **[choice]** where it is a design preference that could go the other way. Every major decision carries an *Alternatives kept in play* block — I prune only what is genuinely dominated.

1. Executive summary

The model I would build. A **program-wise (per-Hallmark) hierarchical Bayesian occupancy-regression**. For each of the 50 Hallmark sets, learn a non-negative repression weight for each candidate (arm → gene) edge from tumour mRNA under a confounder block, with:

- an **occupancy (saturating) link** for how arm abundance turns into repression — not a linear $a \cdot w$ — so the $a \cdot w$ gauge, the RISC-loading threshold, and ceRNA/pool competition are represented by the *functional form* rather than by a hand-set denominator;
- a **spike-and-slab, evidence-graded inclusion prior** $\pi_{\{g,m\}}$ plus an informative **magnitude prior mean** $\mu_{\{g,m\}}$ and **prior strength** $\tau_{\{g,m\}}$, fused from a re-graded evidence stack (curated MTI tiers + **TarBase-v9 method/confidence** + a **single PMID-deduped AGO-CLIP axis** (ENCORIUPSTAR3UTarBase-CLIP; **primary**) + **chimeric duplex coordinates (confirmatory)** + **TargetScan context++ / biochemical K_D + breast literature**), with **site-level** evidence setting magnitude and interaction support and **edge-level** evidence setting inclusion and sign;
- **seed-family partial pooling** so the identified estimand is **family→gene**, with the per-arm split reported *with its posterior width* rather than faked;
- **primed cooperative product terms** only for arm-pairs with distinct sites 8–40 nt apart, and a **shared free-AGO pool** as the mechanism for global saturation / ceRNA;
- a **nested state model** $M_{\text{tumour}} = M_{\text{healthy}} + \Delta$ (healthy GTEx fit as the prior for the tumour fit) that separates *acquired abundance* from *acquired per-site weight*.

Observation target Y. Primary = tumour mRNA level with an explicit transcription-control block in C — a **minimal, composition-robust** set: purity = CPE/ESTIMATE, target-CN, **one malignant-compartment (Cancer-Epithelial CIBERSORTx) proliferation**, a transcription proxy, plus **deconvolution compartment fractions** for stroma/immune programs; HRD and batch off by default (Decision B) — because mRNA carries the majority of miRNA repression at steady state **[assert: Guo 2010; Eichhorn 2014]** and is where the curated evidence and the power are. **Validation Y = CPTAC protein and protein-vs-mRNA discordance** (the translational residual the mRNA fit cannot see) plus an independent mRNA cohort (SCAN-B / METABRIC / Buffa).

Prior scheme. Non-circular by construction: π , μ , τ and the interaction support come from **sequence + curation + external-tissue assays only**, never from TCGA (X,Y); expression

estimates *magnitudes*; the learned-M and curated-M are kept side by side and disagreements are reported as findings.

Evaluation. A falsification ladder with pre-named criteria, scored against the *right* reference per claim: (1) beat **raw abundance** out-of-fold on held-out patients — the primary bar; (2) match or extend **curated fixed-M** while nesting it; (3) beat a **shuffled-evidence null**; (4) a **miRNA-locus CN instrument** (2SLS) for causality; (5) an **orthogonal protein layer** (CP-TAC) and **independent cohort** for OOD. Abandon the learned model for raw abundance if it cannot clear Bar 1.

The 3-5 decisions that matter most. 1. **Observation model + a minimal, composition-robust C** — regress mRNA with a *transcription-control block*, not naked level; validate on protein discordance. Keep C lean: purity, target-CN, **one CIBERSORTx-deconvolved (malignant-compartment) proliferation** term, transcription proxy; drop HRD and batch unless a diagnostic justifies them (over-adjustment/collider risk). (Everything downstream is biased without the transcription control; it is the biggest hole in the current heuristic.) 2. **Occupancy link + abundance-threshold gauge** — fix abundance on the $\log_2(\text{RPM}+1)$ scale, put a saturating gate with a functional-copy-number threshold, let M carry per-site affinity. This resolves the a-w non-identifiability and *retires* $D(m)$ (promiscuity becomes a consequence of a shared finite RISC pool, not an input). 3. **Evidence fusion — de-duplicate by PMID globally (functional AND physical), which is not done today.** The current score counts miRTarBase studies + *adds* ENCORI CLIP on top with **no cross-database dedup**; the same paper double-counts both as CLIP (ENCORI/POSTAR3/TarBase — TarBase is ~93% CLIP from just 998 PMIDs) **and** as functional (a luciferase/western paper curated by both miRTarBase and TarBase). Fix = a unified (**m,g,PMID,assay-class**) **ledger built at edge-build time**, counting *distinct PMIDs* per class. TarBase’s regulation is ~all-Negative (301 Positive in 4.72M) → **not** a direction prior. Sequence/K_D is a separate orthogonal (PMID-free) magnitude channel; K_D gives the **orphan** magnitude. 4. **Family partial pooling** — the identified unit is the seed family; per-arm resolution is a nomination with uncertainty, not a coefficient. 5. **Nested state model** — test whether M itself moves H→tumour (APA/AGO capacity) versus only x moving; this is falsifiable and it decides §7.

2. Literature synthesis (refreshed from the web, July 2026)

What the current state of the art establishes across the §1j facets, with what I **adopt** vs **improve on**, and contradictions flagged. Citations are listed in §8.

2.1 Silencing mechanism & magnitude

- **mRNA destabilisation dominates at steady state.** By the time substantial repression ensues, mRNA decay explains ~66-90% of miRNA-mediated repression; translational repression is rapid but weak (~10-25% residual) [**assert: Guo et al. 2010; Eichhorn et al. 2014**]. Single-cell work shows a hierarchical target response to miRNA induction (Kim/Bartel line) [**assert**]. → **Adopt:** mRNA is a legitimate **primary** readout. → **Improve on:** the translational-only residual is invisible in mRNA and *only* appears in protein / protein-vs-mRNA discordance — so protein is not a nice-to-have, it is the layer that separates a real post-transcriptional edge from a transcriptional confound (§Decision B, §Decision J).
- **Effector biology.** AGO→TNRC6/GW182→CCR4-NOT/PAN2-3 deadenylation→decapping→decay, plus initiation block [**assert**]. → I do **not** model the effector chain explicitly; its net is the saturating decay-rate multiplier in the occupancy link (McGeary’s $b \approx 1.8$, “bound AGO nearly triples decay”).

- **Fine-tuning / mild effects.** Most targets < 2-fold; buffering, threshold-sharpening; one miRNA hits hundreds of proteins, each mildly **[assert: Selbach 2008; Baek 2008]**. → **Adopt:** low SNR per edge ⇒ borrowing strength (priors, groups, low rank) is mandatory, not optional.

2.2 Quantitative site efficacy & 3'UTR biology

- **Site-type hierarchy** 8mer > 7mer-m8 > 7mer-A1 > 6mer, plus 3'-supplementary pairing **[assert]**.
- **TargetScan context++** sums ~14 features (site type, supplementary pairing, local AU, minimum distance, 3'UTR length, SPS, TA, PCT, ORF length/8mer, offset-6mer) into a continuous per-site score **[assert: Agarwal et al. 2015]**. → **Adopt** as a per-edge continuous magnitude prior available for **every** predicted edge, orphans included.
- **The biochemical / K_D model is the current frontier and it changes my recommendation.** AGO-RBNS measures relative K_D for all ≤12-nt sequences; a Langmuir **occupancy** model $N \propto a/(a+K_D)$ predicts repression with $r^2 \approx 0.30-0.37$ (vs ≤0.14 for 30 prior algorithms), and a CNN generalises K_D to arbitrary miRNAs from the 5' sequence (~+50% over TargetScan on held-out miRNAs) **[assert: McGeary, Bartel et al. 2019; scanMiR / Soutschek 2022 packages it]**. Two features I lean on hard: (i) **flanking-dinucleotide AU context spans a ~100-fold affinity range** — often larger than the site-type difference — so “the site exists” is a weak prior; the *graded* K_D is the strong one; (ii) repression scales with **occupancy**, a saturating function of **free** AGO-miRNA concentration and K_D. → **Adopt** the occupancy link as the functional form (§Decision C) and K_D as the mechanistic magnitude prior (§Decision E). → **Improve on:** McGeary fits the free-AGO scalar *a_g* per transfection and holds it fixed across cell types — i.e. the model **openly does not resolve the relative-vs-absolute abundance gauge**; that unresolved scalar is our §3f gauge problem, and I anchor it with the functional threshold rather than leaving it free.
- **Positional / accessibility / APA.** Functional sites enrich near UTR ends, deplete ~15 nt after the stop and in long-UTR interiors; open (accessible) structure predicts efficacy; APA shortens 3'UTRs in proliferating/tumour cells and deletes distal sites **[assert]**. → **Adopt:** APA makes **support itself sample/state-dependent** — a direct argument for the state-dependent M (§Decision H) and for conditioning site presence on 3'UTR isoform where PDUI is available.
- **Deep-learning site predictors** (miRAW, DeepMirTar, miTAR, TargetNet, TEC-miTarget; and 2026 multi-instance models e.g. PAIR-Former) reach high *classification* AUC on curated positives/negatives **[assert]**. → I **do not** adopt these as the estimator: they predict *binary functionality* on curated sets, tend to overfit the negatives, and do not give a calibrated *magnitude* on the natural edge distribution. I adopt only the **biochemical/K_D** line, which is quantitatively grounded and regression-calibrated.

2.3 Seed families, cooperativity & competition

- **Cooperativity window 8-40 nt** between *distinct* sites gives super-additive repression, strongest ~15-26 nt, lost beyond ~40 nt **[assert: Saetrom 2007; Grimson 2007; Broderick 2011]**; the human transcriptome is enriched for cooperativity-permitting spacings **[assert: Rinck/Saetrom-line 2013]**. → **Adopt** as the trigger for *primed* product terms (§Decision G).
- **Seed-family redundancy** — same-seed members are near-substitutable at a shared site (sub-additive), the opposite sign to cooperativity **[assert]**. → **Adopt** as the biological mandate for the family group structure (§Decision F).
- **ceRNA is real but narrow.** Physiological competition requires near-equimolarity / low target abundance; Denzler showed target relief only above $\sim 1.5 \times 10^5$ added sites/cell, insensitive to miRNA level; Bosson (AGO iCLIP) found most active miRNAs are *not*

ceRNA-susceptible, with low-abundance families (miR-25) the exception **[assert: Denzler 2014/2016; Bosson 2014]**. → **Adopt**: model ceRNA/global saturation as a **cohort/sample-level shared-pool** effect (rare, global), **not** a per-edge ceRNA term — the literature says a coarse resolution is the honest one.

- **Functional dose threshold**. Below a per-cell copy floor a miRNA is inert regardless of site quality; a few most-abundant miRNAs dominate AGO loading **[assert: Mullokandov 2012]**. → **Adopt** as the anchor that *breaks the a-w gauge degeneracy* (§Decision C).

2.4 Expression-based regulatory modelling

- **GenMiR++** (Bayesian, sequence-primed, marginal associations) and **TaLasso** (penalised regression, *conditional/partial* associations) are the canonical expression-network methods; sparse network-regularised NMF (**SNMNMf**) and **PIMiM** find miRNA-gene **comodules**; Bayesian **sparse reduced-rank regression** simultaneously does rank reduction + predictor/response selection in $p \gg n$ omics **[assert]**. → **Adopt** the conditional-not-marginal stance (TaLasso), the comodule/low-rank idea (as an *overlay*, §Decision A), and Bayesian sparse-RRR machinery for the universe layer. → **Improve on (the differentiators)**: none of these push a **quality-weighted, assay-directness-graded, breast-audited magnitude** prior into penalty+mean (they use sequence only for *support*); none carry **purity/proliferation/target-CN/transcription** confounder control (composition confounding is the #1 failure mode); none carry an **internal instrument**; none model **cooperativity as primed products**. These four gaps are exactly what our data lets us close.
- **MoPC** (partial correlation of mRNA-protein *conditioned on miRNA*) infers miRNA-target interactions from TCGA/CPTAC multi-omics **[assert, 2024]**. → **Adopt its logic as a validation estimator** on the protein layer (it is precisely the protein-discordance test).

2.5 Causal / identification

- **Invariant Causal Prediction using PAM50 subtypes as “environments”** finds miRNA→mRNA edges whose relationship is *invariant across subtypes*, arguing this is robust to TF confounding; validated against transfection ($|\log_2\text{FC}| > 0.3$); it explicitly **misses subtype-specific edges** **[assert: Meng et al. 2019]**. → **Adopt** the environments idea for the state/subtype axis (§Decision H) and the invariance-vs-specificity tradeoff it names. → **Improve on**: ICP tests invariance but doesn’t estimate a *magnitude* under an evidence prior or carry a dose instrument.
- **miRNA Mendelian randomisation** studies use germline *eQTLs* of circulating miRNAs as instruments for cancer risk **[assert, many 2024-2025]**. → I flag this as a **different design** from ours: those are germline-eQTL, cross-individual, risk-outcome MRs. Our instrument is **somatic miRNA-locus CN perturbing arm dose within tumours** — a within-cohort dose instrument for the *target*, not a risk MR. The MR literature validates the *logic* (genetic dose instrument breaks reverse causation); the *estimand* is ours to define.
- **Composition/purity confounding** in tumour omics is well documented; correlation $\neq$ targeting without adjustment **[assert]**. → **Adopt** into the design C block.

2.6 Protein-level & discordance

- pSILAC established mild, widespread protein effects **[assert: Selbach/Baek 2008]**; CPTAC breast proteogenomics shows real mRNA-protein discordance (e.g. TP53 $r \approx 0.36$; post-transcriptional control of specific genes) **[assert]**. → **Adopt**: protein and discordance are the orthogonal OOD layer; CPTAC phospho/acetyl (now inventoried) give *effector* readouts for select programs.

2.7 State / field-effect biology

- Histologically normal tumour-adjacent breast tissue carries field-cancerisation defects (DNA-methylation drift, CNAs in ~40% of adjacent tissue, altered expression) **[assert]**. → **Adopt the brief's stance as literature-backed: NAT is its own state, never a healthy stand-in**; GTEx is the only bona-fide healthy anchor. Using NAT as normal understates tumour change and can invert directions.

2.8 Breast-specific & Hallmark miRNA biology

- Subtype-specific miRNA programs: miR-200 family high in luminal/epithelial, lost in TNBC (EMT); let-7 high in LumA; miR-21/210/221 up and miR-10b/145/205 down in TNBC; **AGO2-PAR-CLIP shows the target regulation itself differs by subtype**, not just miRNA levels **[assert]**. → **Adopt**: the model is inherently cohort/subtype-specific; subtype is a legitimate "environment" (ties to §Decision H and ICP). The §1h arm/target table is consistent with current literature and I use it as the **known-driver recovery** panel (face validity), not as fitted priors on top of the sequence evidence.

2.9 Evaluation & overfitting

- Sequence target predictors over-predict by ≥ 7 -fold; CLIP shows binding, not function, and carries its own false positives; robust evaluation uses held-out CV, orthogonal-layer and perturbation validation **[assert: Pinzón 2017; Kern 2021]**. → **Adopt**: never trust in-sample fit; every promotion gate is a pre-named held-out / orthogonal win (§Decision J, §6).

Net contradiction check. Nothing in the refreshed literature contradicts the §1 biology or §2 data assumptions. The one *sharpening*: "site exists" is a much weaker prior than the brief's site-type hierarchy alone implies — the **graded K_D (with flanking context)** is the strong prior, and the biochemical model's *unresolved free-AGO scalar* is a candid admission that the abundance gauge (§3f) is genuinely unidentified from expression alone and must be anchored by external biology (the threshold).

3. Decision log

Each entry: **Decision / How / Why / Alternatives kept in play (and when each wins) / Assumptions & risks (and how to detect failure).**

Decision A — Model object (§3a)

Decision. Primary estimator = **program-wise (per-Hallmark) hierarchical Bayesian regression** with a spike-and-slab + informative-normal prior on non-negative weights, an **occupancy link** for the abundance→repression map, seed-family partial pooling, an unpenalised confounder block, and primed cooperative terms. Ship a **frequentist surrogate** (non-negative adaptive group-elastic-net) as the cheap MVP. Keep **low-rank RRR/NMF as a structural overlay** and a **feature/FM model as the orphan prior-mean generator**, not as the primary edge estimator.

How. For a program P with genes G_P and candidate regulators A_P, fit the generative model of §4 jointly over G_P (borrowing strength within the program), with $S \approx 1,060$ tumour samples (primary tumour, PAM50 **Normal-like excluded**; §4.1). Within a program $|G_P| \approx 50-200$, $|A_P|$ a few hundred candidate arms → still $p > n$, made identifiable by sparse+group+informative priors. Estimation by mean-field VI (fast) or Gibbs with a spike-and-slab (gold), each gene's confounder block unpenalised.

Why. - *Biology (§1c, §1h)*: the Hallmark program is the native unit; a program-scoped M reads as “the regulatory matrix of EMT / P53”, ties to the architecture layer, and keeps $p \gg n$ tractable. - *Data (§2a)*: $S \approx 1,060$ (primary tumour, Normal-like excluded), $A \approx 2,236$ (detectable far fewer), $G \approx 4,384$. A single universe fit is $p \gg n$ at its worst and maximally circular; per-program scoping cuts p an order of magnitude while staying interpretable. - *Identifiability*: the informative prior + non-negativity + family groups convert the ill-posed $p \gg n$ problem into a well-posed MAP/posterior; the Bayesian form is the natural home for the multi-source prior (Decision E) and, crucially, **reports uncertainty** so the family non-identifiability (Decision F) is surfaced, not hidden.

Alternatives kept in play. - **Per-gene adaptive-lasso (gene-focused).** Wins as the MVP and for deep single-driver dives (PTEN ~91 arms, GATA3, ESR1, ZEB1) where you want one clean penalised regression, maximum interpretability, and direct comparison to the existing per-edge coupling. Loses cross-gene strength; if orphan support is opened it becomes $p \gg n$ per gene. Use it first; it is Phase 1. - **Low-rank reduced-rank regression / NMF (co-modules).** Wins when regulation is genuinely low-dimensional (a few latent programs) and you want a structural roll-up / borrow maximal strength; it is the SNMNMF/PIMiM/Bayesian-sparse-RRR family. Loses per-edge interpretability (a low-rank M is dense and fights the sparse curated prior). Keep it as an **overlay** that explains residual covariance, not as the edge estimator. - **Feature / factorization-machine model $M_{\{g,m\}} = \phi(g)^T \theta \psi(m)$.** Wins for **reaching orphan edges** — it learns a sequence/evidence $\rightarrow$ weight map that generalises beyond curated support. Loses identification (indirect) and is the most circular (it re-learns the sequence prior). Use it to **generate μ for orphans**, feeding Decision E, not as the primary. - **Mechanistic occupancy GLM (fully nonlinear).** Wins for maximal mechanism fidelity. Loses scalability and needs K_D for canonical sites only. Adopt its **link** inside the regression rather than fitting the full nonlinear system everywhere.

What I would not do. A single universe-wide **dense** RRR as the primary estimator (identifiability bites hardest, circularity highest, interpretability lowest), or a **deep-learning target predictor retrained on expression** (overfits, uninterpretable, and just relearns the sequence prior we already have, per §2.2/§2.9).

Assumptions & risks. Assumes within-program regulation is approximately additive-in-occupancy with sparse support. Fails if a few dense latent factors dominate (detect: large residual covariance after the sparse fit $\rightarrow$ promote the RRR overlay) or if programs are too small for stable fits (detect: posterior inclusion probabilities never separate from the prior $\rightarrow$ fall back to gene-focused or pool programs).

Decision B — Observation target Y (§6) [most load-bearing]

Decision. Primary Y = **tumour mRNA level, regressed with an explicit transcription-control block in C** (not naked level). **Validation Y = CPTAC protein and protein-vs-mRNA discordance**, plus an independent mRNA cohort (SCAN-B / METABRIC / Buffa). Use **mRNA Δ vs a subtype centroid** as the primary only in the state-differenced variant (Decision H), never Δ -vs-NAT.

How. Model $Y_{\{g,s\}} = \alpha_g + C_s \beta_g - \rho_{\{g,s\}} + \epsilon$, where ρ is the repression aggregate (§4). **C should be the minimal set of common causes, not a kitchen sink** (feedback, 2026-07-04): over-adjustment inflates variance and, worse, risks **collider / over-control bias** if a covariate is a *consequence* of the miRNA's effect rather than a cause of both arm and target. Reasoning through each candidate against “is it a plausible common cause of *this* (arm $\rightarrow$ gene) pair?”: - **Purity — keep, but THINK about which purity** (feedback, 2026-07-04). CPE (Consensus Purity Estimate) and ABSOLUTE are **CN/methylation/multi-modal** — largely *orthogonal to expression*. The user's own **breast single-cell-atlas CIBERSORTx** run yields a

Cancer-Epithelial fraction (output/brca_deconvolution/tcga_cibersortx_fractions.tsv) = an *expression-derived* malignant purity. For a confounder in an **mRNA** regression this matters: an expression-derived purity **shares noise with the outcome Y** (both are linear combinations of the same expression), so using it as the purity term risks **partial collinearity / over-control**. → **Decision (2026-07-04, user): use CPE / ESTIMATE for now** — the established purity scalar (CPE is multi-modal and preferred; ESTIMATE is expression-based but on immune/stromal signature genes largely *disjoint* from the target panel, so its circularity is mild). The deconvolution Cancer-Epithelial fraction is kept as a *composition* covariate and a robustness variant, **not** the purity term. - **Target-gene CN — keep (target-side)**. CN drives target mRNA directly and independently of the miRNA; omitting it inflates apparent repression on deleted genes. - **Proliferation — keep exactly ONE, and make it malignant-compartment** (feedback). Proliferation co-drives oncomiRs and their targets via the E2F/MYC axis, but the raw E2F/G2M bulk metagene is itself confounded with tumour-cell fraction (it rises with purity, not only with per-cell cycling). *Correction to my first draft*: the clinical table's cibersortx_* columns are **IFNG-only (LM22, immune)** — a non-immune *residual*, **not** a purified malignant signal. The right tool is the user's **breast single-cell-atlas CIBERSORTx** (cell types incl. **Cancer Epithelial**): compute proliferation on the **Cancer-Epithelial HiRes GEP** (imputed malignant-cell expression — the README's "phase 2" custom epithelial signature), or, as the immediately-available approximation, **residualise bulk proliferation on the Cancer-Epithelial fraction**. Keep thornsson_proliferation / rppa_panel_proliferation and the three E2F/G2M/MKI67 metagenes only as *sensitivity* variants (collinear — do **not** stack). *Caveat*: proliferation is partly *downstream* of the E2F/G2M/MYC Hallmarks, so for those programs conditioning on it risks over-control — report with and without it there. - **Composition — use tissue-matched deconvolution compartment fractions, and they are large-effect for stromal programs** (feedback + repo evidence). Replace the generic ESTIMATE Immune+Stromal / LM22 with the **breast single-cell-atlas deconvolution fractions** the user built — CIBERSORTx (Cancer Epithelial, Normal Epithelial, CAFs, PVL, Endothelial, B/T/Myeloid/Plasmablasts) or BayesPrism (MH-70). The repo's own **coupling deconvolution retest** (core_coupling_deconv_retest, 2026-06-26) shows this is the **gold standard and that the metagene under-corrects**: with 8 cell fractions + proliferation, 732/1,013 headline edges stay FDR-negative (median ρ $-0.148 \rightarrow -0.109$), but the **most-attenuated edges are all miR-29b→ECM/collagen** (COL5A1/2/3, COL3A1, FBN1, SPARC, LOXL2, ADAM12, PDGFRB: $-0.3/-0.4 \rightarrow \approx 0$) — the composition confound made concrete (miR-29 and its ECM targets both track CAF/stromal fraction). So compartment fractions are **essential and program-conditional**: mandatory for stroma/immune-expressed programs (EMT/ECM, IFN, INFLAMMATORY, ANGIOGENESIS); for cell-intrinsic programs (E2F, G2M, MYC, DNA-repair) purity + one malignant-compartment proliferation term usually suffice. - **HRD — drop from the default (reconsidered)**. HRD is a genomic-instability score that is largely a *correlate* of subtype/purity/proliferation, not an independent common cause of a specific (arm→gene) pair; once purity + proliferation + PAM50 are in, it is near-redundant. It sits in the current spine (CONFOUNDER_NUMERIC = CPE + thornsson_hrd_score) mostly by inertia. Keep it only where an ablation shows it moves the estimate. - **Batch — drop unless diagnosed**. TCGA-BRCA miRNA-seq is single-platform (BCGSC); include the plate factor (plate_both) only if a batch diagnostic (guided-PCA / PVCA on the arms) shows a real effect. - **Transcription proxy — keep (the new, essential piece)**. Not currently in the repo covariates; in decreasing cleanliness: (i) **intronic / pre-mRNA read signal** for the target (direct transcription- rate proxy if derivable from STAR), (ii) **TF-regulon activity** for the gene's known TFs, (iii) **latent expression factors** (PEER/PCA) soaking up global transcriptional programs.

So the **minimal default C** = {purity = CPE/ESTIMATE (decided), target-CN, one malignant-compartment proliferation (Cancer-Epithelial CIBERSORTx GEP), transcription proxy}, plus **tissue-matched deconvolution compartment fractions** for

stroma/immune-expressed programs; HRD and batch off by default. Validation: refit-free scoring on CPTAC protein + a **MoPC-style** partial correlation of mRNA-protein *conditioned on the arm* to isolate the translational residual.

Why. - *Mechanism (§1b)*: mRNA carries 66-90% of steady-state repression [**assert: Guo 2010; Eichhorn 2014**] → mRNA is the highest-powered readout and where the curated evidence lives. But miRNA acts post-transcriptionally: target = transcription – repression – other. Regressing on **level** makes transcription a confounder — a TF co-driving a miRNA and its target manufactures a spurious *positive* weight (or masks a real negative one). This is the single largest defect in the current heuristic, which regresses/couples on level with only purity+HRD+prolif in C and **no transcription proxy**. - *The matrix view's gift (§2 of the earlier note)*: the confounder that drives both x and y creates omitted-variable bias in M unless it is in the design. Putting transcription in C is the fix, and it is *cheaper and more honest* than differencing. - *The translational residual is only visible in protein* [**assert: §1b, Selbach**] → protein and especially **discordance** (mRNA present, protein down) is the layer that certifies an edge is truly post-transcriptional and not a surviving transcriptional confound. This makes protein the decisive non-circular OOD check, not a secondary nicety.

Alternatives kept in play. - **Naked mRNA level (no transcription proxy)**. Wins only on simplicity/coverage. Loses to confounding — dominated for causal claims. Keep only as an ablation to *quantify* the transcription confound (compare weights with/without the proxy; the delta is the confound size). - **mRNA residualised on the transcription proxy (fit on residuals)**. Wins when you trust the proxy and want the post-transcriptional part isolated *before* fitting. Loses if the proxy is imperfect (over-residualising removes real signal). Near-equivalent to putting the proxy in C; I prefer in-C because it estimates the adjustment jointly rather than committing to a fixed residualisation. - **mRNA Δ vs reference (GTEx / subtype centroid)**. Wins paired with the state-differenced M (Decision H): gene-specific baseline transcription cancels, improving identification. Loses power (differencing inflates noise) and, if referenced to **NAT**, injects field-effect contamination [**assert: §2.7**] — so reference to GTEx (cross-platform noise cost) or subtype centroid (platform constant, preferred) only. - **Protein level as primary**. Wins if the question is function-first and CPTAC coverage suffices. Loses on n (smaller), MS batch, and that it *understates* the ~75% of repression realised at mRNA. Keep as the co-primary for the subset of genes with strong CPTAC coverage and as the validation layer. - **Protein-vs-mRNA discordance as primary**. Wins as the cleanest post-transcriptional signal. Loses on n and needs paired CPTAC. Strongest as the **validation** estimand (Decision J, Bar 5).

Assumptions & risks. Assumes a usable transcription proxy exists. *Fails* if no proxy captures the relevant TF programs (detect: learned positive weights on canonical repressive edges → proxy inadequate; escalate to differencing or protein-primary). Assumes the mRNA↔protein map is informative; *fails* for translational-only targets (detect: edge couples on protein/discordance but not mRNA → a real edge the mRNA readout structurally misses; flag as translational-class).

Decision C — The a·w scale gauge (§3f) [must-resolve]

Decision. Fix arm abundance on the $\log_2(\text{RPM}+1)$ scale; use a saturating occupancy link with a functional-copy-number threshold; let M carry per-site affinity (1/K). Report **two** quantities, never one: the **potential weight** $M_{\{g,m\}}$ (per-unit affinity, gauge-anchored) and the **realised contribution** $w_{\text{eff}}(m,g,s) = \text{gate}(x_{\{m,s\}}) \cdot M_{\{g,m\}}$ (abundance-gated). The functional threshold — a physical, non-rescalable anchor — is what breaks the degeneracy.

How. The contribution is multiplicatively non-identifiable in the *linear* form a·w (rescale

$x_m \rightarrow c \cdot x_m$, $M \rightarrow M/c$, identical Y). Note this is **not** cured merely by going saturating: $N=a/(a+K)$ is itself invariant to $(a,K) \rightarrow (ca, cK)$ — the *ratio* a/K is identified, not a and K separately. What pins the gauge is **external biology in absolute units**: 1. **Abundance scale fixed** at $\log_2(\text{RPM}+1)$ (no per-arm variance standardisation), so a rare arm and an abundant arm are *not* equalised — the abundant arm loads more RISC, as biology demands (§1e). 2. **A saturating gate with an absolute threshold** $\text{gate}(x) = (x-x_0)_+ / (\kappa + (x-x_0)_+)$ -style (or the biochemical Langmuir occupancy), with the half-max / floor x_0 anchored to the **functional copy-number threshold** (~hundreds of copies; [assert: Mullokandov 2012]) mapped onto the RPM scale. Below x_0 , contribution ≈ 0 *regardless of site quality* — the biology that makes the gauge physical rather than conventional. So “a strong edge” = **high per-site affinity** $1/K$ *among arms that clear the loading threshold*; a rare high-affinity arm has large M but ~ 0 realised repression.

Why. - *Mechanism (§1e; §2.2)*: occupancy is the biochemically correct map from abundance to repression [assert: McGeary 2019], and the free-AGO scalar that McGeary leaves unfit is our gauge freedom — anchoring it with the Mullokandov threshold is the principled resolution. - *It retires $D(m)$* : the promiscuity denominator was a hand-set proxy for “finite RISC spread over many targets.” With a **shared free-AGO pool** (Decision G), a promiscuous arm mechanically dilutes its own per-target occupancy — promiscuity becomes a *consequence*, not an input (see Decision G). - *It aligns the gauge with the question (coupling-invariance lemma, repo EDGE_QUESTION_TAXONOMY)*: for a **single edge**, Spearman coupling is invariant to any monotone edge weight and to the gauge — so the gauge only matters for **attribution/identity** (who owns the gene) and **aggregate magnitude** (how much total). Fixing it once and exposing potential-vs-realised serves both without contaminating the coupling test.

Alternatives kept in play. - **Per-arm variance standardisation (z-scoring x)**. Wins when you only care about *dynamics* (“who moves across tumours”) and want scale-free coupling; it is the current z-spine. Loses by erasing cross-arm abundance (a rare and an abundant arm both get $\text{SD}=1$) — biologically wrong for loading, and rank-invisible to the threshold. Keep for the *coupling* readout, not for attribution or magnitude. - **Row/column-norm constraint on M** ($\|M_{\cdot, m}\|=1$). Wins mathematically (removes the degeneracy cleanly). Loses biological meaning (an arbitrary norm). Dominated by the threshold gauge, which is physical. - **Fix abundance, let M carry magnitude, no gate (linear)**. Wins on simplicity and is what the RRR sketch assumed. Loses the threshold and saturation biology; a low-abundance arm with a great site gets over-credited. Keep as the unsaturated-regime approximation ($a \ll K \Rightarrow \text{occ} \approx a/K$) where it is valid.

Assumptions & risks. Assumes RPM is a usable monotone proxy for per-cell copy number up to a per-sample constant. Fails under strong compositional distortion (a global miRNA-content shift rescales all RPMs) — detect via the AGO/RISC-capacity covariate and via CN-instrument concordance; mitigate by the shared-pool normalisation (Decision G) which absorbs per-sample loading. The threshold x_0 is uncertain; treat it as a **hyperparameter tuned out-of-fold**, and report sensitivity.

Decision D — Inclusion / HE rule and hard-vs-soft mechanism (§3b)

Decision. **Soft, evidence-graded spike-and-slab inclusion** as the mechanism, run in **two disciplined modes side by side**: a **hard-HE “confirmatory” model** (trusted backbone) and a **soft “discovery” model** (orphans admitted at low prior, must clear OOD to be believed). **Redefine HE** away from the miRTarBase-only rule by folding in TarBase-v9 method/confidence, a PMID-deduped AGO-CLIP axis (chimeric confirmatory), context++/K_D magnitude, and breast literature (Decision E builds the grade).

How. Inclusion indicator $z_{\{g,m\}} \sim \text{Bernoulli}(\pi_{\{g,m\}})$, π a monotone fusion of the re-graded evidence (Decision E). The confirmatory model sets $\pi=1$ on the HE set and $\pi=0$ else-

where (hard mask); the discovery model sets $\pi \in (0, 1)$ graded, high for strong edges, low-but-nonzero for orphans with real sequence/physical support — so a weak edge enters **only if the expression insists** while the prior holds the noise floor down. The two are the same generative model at two prior settings, so they are directly comparable and the discovery model **neests** the confirmatory one.

Why. - *It is not fence-sitting — the two answer different $\$0$ questions:* confirmatory = “re-produce curated knowledge more parsimoniously / test it”; discovery = “extend to edges curation never had.” Both references are live per $\$0$; forcing a single mode discards one question. - *Biology ($\$1d$):* orphans often carry strong context++/K_D/chimeric support (60% of our high-value orphans have direct duplex reads [**assert: Manakov, repo memory**]) — hard-restricting throws away the continuous signal and the discovery lane. But orphans are also where false positives live (predictors over-predict ≥ 7 -fold [**assert: Pinzón 2017**]) — so they enter *soft*, gated by OOD, not as trusted coefficients. - *Identifiability:* the hard-HE model is the most identifiable (cleanest support); it is the backbone whose weights you quote. The soft model trades identifiability for discovery under an explicit validation gate.

Alternatives kept in play. - **Hard-restrict only.** Wins when the goal is a defensible, maximally-identifiable curated-weight refinement and you distrust discovery. Loses all orphan discovery and the continuous signal. - **Soft only (no confirmatory backbone).** Wins for maximal discovery. Loses a trusted reference and invites over-fitting the orphan tail. Dominated by running both. - **Grade granularity** is itself a choice: binary HE/not, vs a 3-tier (HE / evidence-tail / orphan), vs a fully continuous π . *Continuous wins* for the soft model (uses all information); *tiered wins* for reporting and for the hard confirmatory mask. Ship continuous π with tier labels for interpretability.

Assumptions & risks. Assumes the evidence grade is calibrated to true functionality. Fails if a source is miscalibrated (e.g. CLIP depth admits binding-not-repression) — detect by the source-reliability calibration in Decision E (does this evidence class predict held-out protein repression?); down-weight the offending source. Spike-and-slab can be unstable across folds for near-threshold edges — report **posterior inclusion probabilities**, not hard in/out, and treat PIP as the discovery score.

Decision E — Multi-source prior fusion ($\$3b/\$3e$) [core differentiator]

Decision. A two-layer evidence model honouring the edge-level vs site-level split: **site-level evidence** → **magnitude prior μ and interaction support**; **edge-level evidence** → **inclusion π , sign, and prior strength τ** . Fuse sources by **calibrated log-odds/log-magnitude addition** with source-reliability weights *learned against held-out functional repression*, not hand-set. Cap data override by τ .

How. - **Site-level (magnitude + interactions).** context++ / biochemical K_D are per-site, continuous, mechanistic, and defined for orphans. Aggregate sites in the gene’s 3’UTR by the **occupancy sum** (not a mean), so multiplicity is encoded: $\mu_{\{g,m\}} = \log(1 + \sum_{\{sites\}} occ_prior(K_{\{s\}}))$, occ_prior from the biochemical Langmuir at a reference abundance. This gives a magnitude prior for **every** predicted edge, orphans included, and is strictly better than the current $\log(1+evidence)$ (which is flat within a gene, max/min $\approx 1.8\times$, and blind to multiplicity). Site **coordinates** prime interaction support: distinct sites 8–40 nt apart → a cooperative product candidate (Decision G); same/overlapping sites → sub-additive grouping. **Which predicted sites are actually occupied** is marked by the *site-level* physical layers — **POSTAR3 AGO2 peaks** (dense), TarBase-v9 CLIP coordinates, and Manakov chimeric duplex coordinates — which up-weight occ_prior on covered sites and confirm the cooperative-pair support. - **Edge-level (inclusion + sign + confidence).** A logit for π : $logit\ \pi_{\{g,m\}} = w_0 + w_{\text{tier}} \cdot \text{tier}(\text{curated MTI: reporter/protein/perturbation} > \text{RNA} > \text{binding})$

+ w_meth.TarBase9_method_conf(PMID-deduped) # NB regulation $\approx$ all-Negative $\rightarrow$ not a direction channel + w_ago.loglp(AGO-CLIP: ENCORIuPOSTAR3uTarBase-CLIP, PMID-deduped) + w_chim.loglp(chimeric duplex reads) + w_lit.loglp(breast PMIDs) + w_seq.seq_support_indicator # POSTAR3 peaks $\rightarrow$ μ (site level). **Sign — with a big caveat** (checked on the live file): TarBase v9’s regulation is **near-degenerate, 4,724,236 Negative vs 301 Positive** — an almost-constant curation default (most rows are CLIP, which measures *binding, not direction*), **not** an informative per-edge sign, so it is *not* the rich directional prior my first draft claimed. Actionable use: route the **~301 explicit Positive** edges to the *activating/indirect* class or exclude them (a validated miRNA-up $\rightarrow$ target-up “edge” must not be forced to $M \geq 0$); **otherwise take sign from genuine functional assays** (miRTarBase functional tiers, luciferase, western) plus the $M \geq 0$ prior — never from TarBase’s blanket “Negative”. - **The two CLIP layers are different tools — lean on AGO-CLIP as primary, and use several AGO-CLIP sources** (feedback, 2026-07-04). **AGO-CLIP (HITS/PAR/eCLIP)** is dense, replicated — but **miRNA-agnostic** (it shows AGO occupies a region; *which arm* is inferred by seed-matching), so it is the **workhorse for region/edge inclusion π** and for marking bound sites in the cooperativity window (Decision G). **Do not rely on ENCORI alone — there are multiple in-repo AGO-CLIP sources, and ENCORI is sparse** (the method_dev site-ladder found ENCORI AGO-CLIP covered only ~3% of predicted sites): - **POSTAR3 carries ~2.36M AGO2 binding-site rows** (data/RBP-RNA/POSTAR3.parquet) — a far denser, uniformly-processed compendium; already in-repo (used for lncRNA-RBP, repurposable here). - **ENCORI/starBase ships edge-level interaction calls** (the current $\alpha=0.5$ boost). - **breast-context AGO2-PAR-CLIP** (Farazi 2014, subtype-resolved) — a **tissue-matched** physical prior and a subtype signal (ties to Decision H); a literature add, not yet in-repo. **Do not double-count — and this critically includes TarBase v9** (feedback, 2026-07-04). Checked against the live file: **TarBase v9 is ~93% AGO-CLIP itself** (HITS-CLIP 2.93M + PAR-CLIP 1.49M of 4.72M human rows) plus chimeric (qCLASH/CLASH/Chimeric-fragments ~0.14M, the **same class as Manakov**), drawn from only **998 distinct PubMed IDs** — so ENCORI, POSTAR3, and TarBase’s CLIP rows are largely the **same underlying experiments**. Therefore treat *physical AGO-CLIP binding as ONE axis* = **union/max over {ENCORI, POSTAR3, TarBase-CLIP rows}**, de-duplicated by **article_pubmed_id**, never a sum; and fold TarBase’s qCLASH/CLASH rows into the **chimeric** axis with Manakov. Following the repo’s site-ladder rule, ENCORI is the edge-level representative (π) and POSTAR3 the site-level representative (μ /occupancy). **TarBase v9’s genuine value-add is NOT “more CLIP”** — it is the thin **functional layer** (Luciferase 3.3k, Western 1.3k, pSILAC 6.5k, qPCR 961 $\rightarrow$ functional-repression tier, PMID-deduped against miRTarBase), plus **confidence, tissue/cell_line, microt_score, and site coordinates**. **Chimeric eCLIP / CLASH (Manakov)** uniquely ligates the miRNA to its target, giving the direct (**arm, site**) **pair** — invaluable for *arm assignment* and interaction priming — **but it is sparse and its reproducibility is contested** (a view in the field is that chimeric maps are hard to replicate and capture many non-canonical, uncertain-function duplexes) [**concern raised; treat as a caveat, not settled**]. Therefore: **use AGO-CLIP depth as the primary physical π signal**; use chimeric only as a **confirmatory arm-assignment / site-coordinate layer where its depth is adequate**, and **fall back to AGO-CLIP region + seed inference when chimeric is too sparse** for an arm/gene. Neither sets sign or large μ on its own — proximity $\neq$ repression (the 0.5 CLIP discount, now mechanistic); functional / directional evidence is still required for magnitude and sign. - **Provenance de-duplication is GLOBAL — one (PMID $\times$ assay-class) counts once, across every database and both the physical and the functional layers, before any fusion** (feedback, 2026-07-04). **This is not done today and is a real gap**: the current evidence_score is built from miRTarBase’s own per-class **study counts** (n_reporter__functional_mti_studies, ...) with the ENCORI AGO-CLIP depth **added on top** ($\alpha=0.5$), and TarBase is not yet wired — so there is **no cross-database PMID dedup**, and the edges table the model consumes (mirna_hallmark_edges.tsv.gz) has **already collapsed to counts with no PMID column**. The same paper therefore double-counts: (i) a **CLIP study** surfaced by ENCORI and POSTAR3 and TarBase (TarBase = 998 PMIDs $\rightarrow$ millions of rows);

(ii) a **functional** paper (a luciferase or western assay) curated by **both miRTarBase and TarBase** — the functional tier double-counts exactly like the CLIP tier. **Fix — build a unified evidence ledger at edge-build time** (upstream of the count-collapse), keyed by (m, g, pmid, assay_class, direction, tissue), drawing PMIDs from miRTarBase References (PMID), TarBase article_pubmed_id, and each CLIP/chimeric study; **dedupe to distinct (m, g, pmid, assay_class)** (dedupe on PMID $\times$ assay, *not* bare PMID — one paper with both a reporter *and* a western contributes two independent assay-classes), then aggregate per edge per class as a **count of distinct supporting PMIDs**, not rows/peaks/DB-hits. Only then apply the tier weights + log1p. Two provisos: (a) the **sequence/biochemical priors** (context++, K_D, TargetScan) have **no PMID and are a separate orthogonal channel** — they are *predictions*, correctly excluded from experimental dedup; (b) where a source exposes no PMID (some ENCORI cache columns $\rightarrow$ only clipExpNum; POSTAR3 $\rightarrow$ dataset IDs), dedupe by the best available experiment/GEO/cell-line identifier and **cap** the DB's own count as an upper bound, flagging the edge as lower-provenance. - **The fusion is METHOD-centric, not source-centric** (the right frame for the model we are building). The evidence axis is the **assay method** — {reporter/luciferase, protein/western, qPCR/RNA, perturbation, AGO-CLIP, chimeric/CLASH, degradome} — and **every database pours its distinct PMIDs into those method bins**; a database is *never* its own additive term. This is the correct structure precisely because it is where the dedup lives (dedup = distinct PMIDs *per method, across sources*). *Diagnosis of today's score*: it is **method-centric within miRTarBase** (its cross-count classes reporter/protein/rna/perturbation/ binding already *are* assay methods) **but source-centric across databases** — ENCORI is bolted on as its own composite + $\alpha \cdot \text{enc_depth}$, so the **AGO-CLIP method is split** (miRTarBase n_binding at $w=0.5$ and ENCORI clipExpNum at $w=2.0 \cdot \alpha$, un-pooled, un-deduped), and worse, enc_depth **mixes sequence predictions** (1[ind][TargetScan\miRanda\PIITA]) into what should be a physical-assay term. The build folds all of these into one method-centric, PMID-deduped set of bins, with predictions moved to the separate sequence/K_D channel (μ). - **Fusion & calibration — the method weights are LEARNED, not hand-set, and not on TCGA coupling**. The per-method contributions combine additively on the logit/log scale (Bayesian evidence combination) **over the de-duplicated distinct-PMID counts** (never raw per-DB counts). The method weights w_{method} (reporter vs western vs CLIP ...), the weak-discount, and the relative binding-vs-functional weight are **fit** — a logistic/Cox regression (or the outer Bayesian model's hyperprior) predicting an **external validated-repression outcome** the coupling test never sees (perturbation $\Delta \log \text{FC}$ from transfection/KO series; a held-out functional-MTI label), by **nested cross-validation, with uncertainty on each weight. This is a real correction to the current pipeline**, whose weights ($w_{\text{rep}}=3.0$, $w_{\text{prot}}=2.5$, $w_{\text{rna}}=1.5$, $w_{\text{bind}}=0.5$, $\delta=0.3$, $\alpha=0.5$) are **hand-set constants** “selected” once by a small grid (dual_spine_comparison) on **TCGA coupling** — i.e. tuned on the very outcome the model is later scored against (**circular**, and it violates the guardrail below). Everything non-prior (λ , τ , x_0 , pool params, low-rank rank) is likewise tuned **out-of-fold**, never in-sample. $\tau_{\{g,m\}}$ (prior strength) decreases with evidence: HE edges get tight τ (data barely moves them); orphans get loose τ (data free to speak, but low π holds the floor). - **Override budget & non-circularity**. How far expression may move an edge is set by τ . All of π, μ, τ and the interaction support come from **sequence + curation + external-tissue assays only** — never TCGA (X, Y) — so the guardrail (§8 of the earlier note) holds.

Why. - *It is the stated differentiator (§1 of the earlier note)*: prior at **magnitude**, not just support; quality-weighted, assay-directness-graded, breast-audited. GenMiR++/LASSO use sequence for support only. - *Data (§2c), corrected against the live file*: TarBase-v9 (4.72M rows) is **~93% AGO-CLIP** (HITS/PAR-CLIP) + chimeric (qCLASH/CLASH) from **998 PMIDs**, with **site coordinates** and microt_score but a **near-degenerate direction** (301 Positive) — so its first-class value is **site coordinates + method breadth/confidence + a thin functional layer**, fused into the *shared* CLIP/chimeric axes (PMID-deduped, not additive to ENCORI/POSTAR3/Manakov), *not* an independent directional layer. Manakov chimeric (direct

duplex, **site-level**) is likewise partly shared with TarBase’s qCLASH. The biochemical K_D gives the mechanistic magnitude for orphans. The genuinely under-used, non-redundant signals are the **site coordinates** (interaction priming) and **K_D magnitude**, not “more CLIP mass”. - *Edge vs site reconciliation*: the split resolves the brief’s hardest fusion question — edge-level assertions set *whether and which sign*, site-level scores set *how much and which interactions*.

Alternatives kept in play. - **Single fused scalar (no edge/site split)**. Wins on simplicity. Loses the ability to prime interactions and to use orphan site magnitudes — dominated for the cooperativity and orphan goals. - **Hand-set source weights (current)**. Wins on transparency and needs no calibration data. Loses calibration; keep as the initialisation and as a robustness check (does calibration move conclusions?). - **Feature/FM model for μ** (Decision A alternative) instead of the occupancy-sum. Wins when you want a *learned* sequence→magnitude map (more expressive than the biochemical prior). Loses circularity headroom. Use it only to fill μ where the biochemical model has no K_D (non-canonical sites).

Assumptions & risks. Assumes sources are conditionally informative and the calibration set is representative. **The dominant risk is source correlation / counting the same paper more than once — in both the CLIP tier and the functional tier**: ENCORI/POSTAR3/TarBase-v9 (~93% HITS/PAR-CLIP, 998 PMIDs) share CLIP experiments, and miRTarBase $\cap$ TarBase share the same curated *functional* papers (luciferase, western). **Mandatorily de-duplicate by (PMID $\times$ assay-class) globally** (the provenance-ledger step above), or the prior fakes independent confirmation — and note this is a change from the current pipeline, which does *not* dedup and *adds* ENCORI on top of miRTarBase counts. TarBase’s regulation is near-constant “Negative”, so it must **not** be used as a directional/inclusion signal beyond the ~301 explicit Positives; carry its *tissue/cell_line* to down-weight non-breast rows, and take sign from functional assays.

Decision F — Family identifiability (§3d)

Decision. Hierarchical partial pooling toward a seed-family latent: $M_{\{g,m\}} = M_{\{g,F(m)\}} + \delta_{\{g,m\}}$, $\delta \sim N(0, \rho_F^2)$. The **identified estimand is family→gene** $M_{\{g,F\}}$; the per-arm split is a **nomination reported with its posterior width**, never a point coefficient. Use the **real seed** (TargetScan miR_Family_Info, positions 2-7/2-8), not the name-stem proxy.

How. Same-seed arms are near-collinear columns of X (within-family $\rho \approx 2\times$ between; hubs 20-37% redundant [**assert: repo**]). Partial pooling makes $M_{\{g,F\}}$ identified (the shared, well-conditioned direction) and δ weakly identified: when members are perfectly collinear, δ ’s posterior = its prior (“cannot split”); when a member’s abundance diverges (arm-switch, differential expression, isomiR), δ becomes identified and the model *can* resolve the member.

Why. - *Biology (§1g)*: seed-family redundancy is *mandated*, not a numerical nuisance — same-seed members hit overlapping sites and are substitutable (sub-additive). The group structure is the sub-additivity model (Decision G row 1). - *Identifiability*: plain LASSO keeps one member arbitrarily and unstably across folds — reporting an unidentified quantity as if identified. Partial pooling reports the ridge honestly (“family carries W ; member $A = 0.6W \pm 0.3$, not distinguishable from B ”) — matching the repo’s identity-vs-magnitude and “per-arm is a nomination” stance.

Alternatives kept in play. - **Group-LASSO (family in/out together)**. Wins as the frequentist middle ground (cheaper than Bayes, gives the group selection). Loses the calibrated within-family uncertainty (still needs a post-hoc split rule). Use in the frequentist surrogate. - **Collapse to one family predictor (mean/max/sum)**. Wins for maximum power and as the MVP (already built: *mir301_family_depth*); the pooled family abundance is the cleanest single predictor. Loses the ability to ever resolve a member. Use when the split is not the question.

- **No grouping (per-arm).** Wins only if members are genuinely independent (distinct seeds mislabelled as a family). Loses to collinearity otherwise — dominated for true families.

Assumptions & risks. Assumes the seed family is the right grouping. Fails for arms whose *isomiR* distribution shifts the seed (5' *isomiR*s change positions 2-7 → a different family) — detect via *isomiR* data (Decision/Open-question); a 5'-heterogeneous arm should be split by *isomiR* before grouping. Also fails if pooling is too strong (masks a real member-specific edge) — detect by posterior-predictive checks on the member with divergent abundance.

Decision G — Competition, cooperativity, ceRNA (§3e)

Decision. Three couplings, three mechanisms, three resolutions: 1. **Seed redundancy (miRNA↔miRNA, same site, sub-additive)** → the **family group structure** (Decision F); optional per-site saturation for overlapping-but-different-seed arms. 2. **Cooperativity (miRNA↔miRNA, distinct sites 8-40 nt, super-additive)** → **primed product terms** $+\psi_{\{g,(m,m')\}} \cdot \text{occ}_{\{m,g\}} \cdot \text{occ}_{\{m',g\}}$, support drawn from TargetScan/TarBase site coordinates, strong spike-and-slab with a low base rate. 3. **ceRNA / global saturation (target↔target, shared finite RISC)** → a **cohort/sample-level shared free-AGO pool**, not a per-edge ceRNA term. This makes the promiscuity denominator $D(m)$ **obsolete**.

How. - *Cooperativity*: only arm-pairs with **distinct predicted sites 8-40 nt apart on the same 3'UTR** get a product term (Saetrom/Grimson/Broderick window [assert]). The prior does double duty: main-effect support (edge-level) + interaction support (site coords). Products are collinear with their mains, so promote a product only if it beats its mains out-of-fold. - *Shared pool (ceRNA + promiscuity)*: free-loading scalar per sample $\phi_s = A_{\text{total},s} / (1 + \sum_{\{m,g\}} \text{occ}_{\{m,g,s\}})$, and realised weight $w_{\text{eff}} = \phi_s \cdot \text{occ}$. A promiscuous arm targeting many sites raises the denominator → **dilutes its own per-target occupancy**; a highly-expressed target sinks occupancy from others through the same pool. This is the target-site-abundance / AGO-depletion mechanism, per-arm and mechanistic, replacing the hand-set $D(m)$.

Why. - *Biology (§1d/§1f)*: cooperativity and redundancy have **opposite signs** and must not be conflated (the earlier note's §4). ceRNA is real but narrow — physiological only near equimolarity/low target abundance [assert: **Denzler 2014/16; Bosson 2014**] — so a **global** shared-pool term is the honest resolution, and a per-edge ceRNA term would over-fit a rare effect. - *Retiring $D(m)$* : promiscuity as a *consequence* of finite-RISC competition is both more principled and testable — the model should reproduce the promiscuity penalty endogenously, and we can check that the fitted per-target occupancy of hub arms is depressed without any $D(m)$.

Alternatives kept in play. - **No interaction terms (pure additive).** Wins for identifiability and as the MVP — additivity is a fine *local* first order (§1c). Loses all cooperativity/saturation. Start here; add terms only if they earn OOF. - **Per-edge ceRNA terms.** Wins if a specific low-abundance family (miR-25-like) shows physiological competition. Loses generally (over-fits a rare effect). Reserve for named low-abundance cases with evidence. - **Keep $D(m)$ as a hand-set denominator.** Wins on simplicity if the shared-pool fit is unstable. Loses principle. Keep as fallback; report whether the pool reproduces $D(m)$'s effect.

Assumptions & risks. Product terms risk over-fitting $(A(A-1)/2)$ candidates — mitigated by strict site-priming + shrinkage + OOF promotion. The shared-pool ϕ_s may be under-identified without an AGO-capacity measurement — anchor it to the RISC-capacity covariate already in the repo (*ago_gate*), and detect failure by whether ϕ_s correlates with measured AGO/TNRC6 levels.

Decision H — State axis: GTEX→NAT→tumour (§7)

Decision. **Nested/hierarchical state model** $M_{\text{tumour}} = M_{\text{healthy}} + \Delta$, shared support, partially-pooled magnitudes, with the **healthy (GTEX) fit as the prior for the tumour fit**. Separate M per state kept as an audit; state-differenced $\Delta Y = -M \cdot \Delta x$ used on the **tumour-vs-subtype-centroid** contrast (platform-constant) where differencing helps identification. **NAT is its own state, never a healthy stand-in.**

How. Fit M_H on GTEX healthy breast; use it as the prior mean for M_T , so M_T shrinks toward M_H and $\Delta = M_T - M_H$ is “what the tumour data insists changed.” This **separates acquired abundance (Δx , the robust cross-state signal) from acquired per-site weight (Δ in M)** — the §1i distinction. Biological prediction: $\Delta \neq 0$ should concentrate where **APA shortens 3’UTRs** (distal site loss → weight drops) and where **AGO/TNRC6/Dicer capacity** shifts (global rescale). Test $\Delta = 0$ as a falsifiable null: if $\Delta = 0$ everywhere, a single M + acquired abundance suffices (a clean result).

Why. - *Gauge (Decision C):* two **separately-gauged** matrices are not comparable — nesting shares the gauge by construction, so Δ is meaningful. This is the decisive argument against separate- M -as-primary. - *Power:* NAT $n=104$, GTEX $n=346$ (cross-platform) are weak for independent fits; nesting borrows the tumour $n \approx 1,060$ to anchor and estimates only the *deviation*. - *ICP precedent (§2.5):* Meng 2019 uses PAM50 subtypes as environments and finds invariance robust to TF confounding but **misses subtype-specific edges**. So run the nested model **and** flag edges whose Δ (or whose weight) is subtype/state-specific as *findings* (state-dependent regulation), not failures — capturing exactly what pooling-for-invariance discards. - *Biology (§1i, §2.7):* APA/site-loss and effector-capacity change make M state-dependent in principle — this must be **tested**, and the nested model is the test.

Alternatives kept in play. - **Separate M per state.** Wins for cleanest per-state interpretation if each state had ample, same-platform n . Loses power and cross-gauge comparability. Audit only. - **Pooled + state interaction (M_0 + state offset).** Wins for power and simplicity. Loses the APA-driven support contraction (assumes shared support). Near-equivalent to nested when Δ is small; prefer nested because it makes Δ explicit and testable. - **State-differenced $\Delta Y = -M \cdot \Delta x$.** Wins on identification (baselines cancel — addresses Decision B’s transcription confound directly). Loses to GTEX cross-platform noise and NAT field-effect. Use on the **tumour-vs-subtype-centroid** contrast (platform constant), pairing with Decision B’s Δ variant. - **Subtype-as-environment instead of tissue-state.** Wins for causal robustness (ICP). Complementary, not exclusive — run subtype environments *within* the tumour state for the invariance check.

Assumptions & risks. Assumes GTEX healthy is a valid prior anchor despite platform differences — mitigated by the existing quantile-normalisation and MIMAT-join machinery, and by the miTED-anchor imputation for GTEX-collapsed arms (already built). *Fails* if GTEX-TCGA platform effects masquerade as Δ — detect by whether Δ concentrates in the GTEX-imputed/low-confidence arms (an artifact signature) vs in APA-shortened genes (a biology signature).

Decision I — Aggregation on a gene, attribution to a miRNA (§4)

Decision. **Aggregate = the fitted model’s predicted repression** (occupancy-summed then saturated), not a raw sum of hand-weighted contributions. **Attribution = expose two distinct objects, never conflated: magnitude/force** = realised contribution $w_{\text{eff}}(m, g, s)$ and its gene sum; **identity** = a **Shapley decomposition of the fitted aggregate** across regulators. **Retire the softmax share.**

How. - *Aggregate:* sum in **occupancy** space (additive loading), then apply the global sat-

uration/pool — $\rho_{\{g,s\}} = \text{sat}(\sum_m w_{\text{eff}} + \sum \text{coop})$. This replaces the current $\text{AGG} = \sum$ contributions. - *Identity (who owns the gene)*: the **Shapley value** of each regulator's contribution to the fitted aggregate — the unique fair split that handles collinearity (redundant same-seed arms *split* credit rather than one taking it all). For the linear-in-occupancy regime this reduces to the **LMG / averaging-over-orderings** R^2 decomposition; for the nonlinear occupancy aggregate use sampling/kernel Shapley. This **replaces the softmax share**, which (a) spends abundance twice (once in the exponential share, once in $\log_{\text{rpm}}$) and (b) is rank-invisible to Spearman — both documented repo defects (FORMULAS.md §5a). The softmax share is *nested* as the degenerate Shapley of a proportional-abundance model, so we can show what it approximated and where it failed. - *Magnitude (how much total)*: w_{eff} and $\sum_m w_{\text{eff}}$ directly — the force/pressure question.

Why. - *The identity≠magnitude trap (repo memory attribution-identity-vs-magnitude)*: a share summing to 1 answers “who,” not “how much.” Coupling/repression claims need **magnitude** + the coupling test; “which arm owns the gene” needs **identity** (Shapley). Report both; enforce the repo discipline that **share ⊥ coupling** (a high identity share is *never* evidence of repression — only the coupling test is; FORMULAS.md §5a caveat carries over to Shapley unchanged). - *Gene↔miRNA spectrum*: all one M. Column of M over g = an arm's total targeting role; row over m = a gene's total incoming pressure (layer impact); a **seed family / polycistron (miR-17~92)** = the group latent (Decision F) or a defined arm-set aggregate — the model exposes these *for free* without extra machinery.

Alternatives kept in play. - **Softmax share (current)**. Wins on cheapness and it is already wired. Loses on the double-counting and rank-invisibility. Keep only as a documented baseline to show the improvement. - **Fitted contribution directly, no Shapley**. Wins when regulators are near-orthogonal (Shapley≈raw contribution) and you want speed. Loses under collinearity (over-credits one of a redundant pair) — which is exactly the family case. Use raw contribution for magnitude, Shapley for identity. - **Sum vs mean aggregation; signed/positive/absolute coherence; pruning; RISC-gating**. These are *reporting* choices on the fitted aggregate. *Sum* wins for total force; *mean* for per-regulator intensity; *signed coherence* as a diagnostic (do regulators agree in direction?). Ship sum as headline, the others as diagnostics, mirroring the current share-metric menu but on the *fitted* model.

Assumptions & risks. Shapley is exponential in regulators — use the linear/LMG closed form where valid and sampling elsewhere; for genes with many regulators, cluster to families first (Decision F) then Shapley over families (also the identified unit). Risk: users read identity as repression — mitigate by never shipping identity without the coupling flag beside it (the repo's driver_candidate = coupled AND high share discipline).

Decision J — Evaluation & references (§5, §0)

Decision. A **falsification ladder** with pre-named criteria, each scored against the *right* reference per §0, plus pre-named **abandon** conditions. Identification lives in the *fit* (the C block and the instrument), not only in the checks — Q1 and Q3 are one problem.

(Full test specs, success thresholds, and abandon conditions in §5.)

Why. The references are not interchangeable (§0): sometimes the bar is **raw abundance** (does structure beat just-abundance?), sometimes **curated fixed-M** (does learning beat/parsimoniously match hand weights?), sometimes a **shuffled-evidence null** (is the prior doing work?). The **coupling- invariance lemma** (repo) forces a key design point: a *single-edge* Spearman coupling **cannot** distinguish edge weightings (it is invariant to monotone edge_w and to the gauge) — so to show learned weights beat curated ones you **must** test at the **aggregate** (multi-regulator) level or by **out-of-fold prediction**, never by single-edge coupling. This is why Bars 1-2 are aggregate/predictive.

Alternatives kept in play. Which reference is “the bar” depends on the claim being made — keep all three live and report against each, rather than declaring “beat the heuristic” the sole goal (the brief is explicit that it is not).

Assumptions & risks. The instrument may be weak (many loci have little CN variance) — report first-stage F and restrict to focal segments; if no arm has an adequate instrument, the causal bar is *inconclusive*, not *failed*. Orthogonal-layer coverage (CPTAC) is partial — pre-register the gene set with adequate protein coverage.

4. Model specification — recommended default (and variants)

Every symbol defined. This is precise enough to implement.

4.1 Indices, data, gauge

- $s \in \{1..S\}$ samples (tumour $S \approx 1,060$ — primary tumour with PAM50 **Normal-like excluded**, $n=36$ dropped via `config.EXCLUDE_NORMAL_LIKE`, from the $\sim 1,096$ primary-tumour miRNA set; NAT 104; GTEx 346; metastatic 7 — **dropped**, too few).
- $m \in A_P$ candidate arms for program P; $g \in G_P$ genes; $F(m)$ = seed family of m.
- $X_{\{s,m\}} = \log_2(\text{RPM}_{\{s,m\}} + 1)$ — **the fixed abundance gauge** (Decision C); no per-arm variance standardisation for the magnitude/attribution model.
- $Y_{\{s,g\}}$ = primary readout = tumour mRNA $\log_2(\text{TPM} + 1)$ (Decision B).
- $C_{\{s,\cdot\}}$ = **minimal-core** confounders (Decision B): **purity = CPE/ESTIMATE** (decided 2026-07-04; not the expression-derived Cancer-Epithelial fraction), target-gene CN, **one malignant-compartment proliferation** (Cancer-Epithelial CIBERSORTx HiRes GEP; the E2F/G2M/MKI67 metagenes and thornsson_proliferation as *sensitivity only* — not stacked), and a **transcription proxy** (intronic/pre-mRNA or TF-regulon or latent factors). **Program-conditional** add-ons for stroma/immune programs: the **breast single-cell-atlas deconvolution fractions** (CIBERSORTx Cancer/Normal Epithelial, CAFs, PVL, Endothelial, immune; or BayesPrism) — gold-standard vs the metagene (retest). **HRD and batch are off by default** (near-redundant / single-platform; include only if an ablation or batch diagnostic justifies them).

4.2 Occupancy link (abundance $\rightarrow$ repression), gauge-anchored

For arm m, gene g, sample s, with per-site effective dissociation $\kappa_{\{g,m\}}$ (low = strong site):

$$\text{occ}_{\{m,g,s\}} = a_{\{m,s\}} / (a_{\{m,s\}} + \kappa_{\{g,m\}}), \quad a_{\{m,s\}} = (X_{\{s,m\}} - x_0)_+ \quad \# \text{ threshold-gated abundance}$$

x_0 = functional-threshold floor on the RPM scale [**assert: Mullokandov 2012**], tuned out-of-fold. Unsaturated approximation ($a \ll \kappa$, the MVP): $\text{occ} \approx a_{\{m,s\}} \cdot \theta_{\{g,m\}}$, $\theta_{\{g,m\}} = 1/\kappa_{\{g,m\}}$ = the learned non-negative weight $M_{\{g,m\}}$.

Shared free-AGO pool (Decision G; ceRNA + promiscuity), per sample:

$$\phi_s = A_{\text{total},s} / (1 + \sum_{\{m,g\}} \text{occ}_{\{m,g,s\}}), \quad w_{\text{eff}}(m,g,s) = \phi_s \cdot \text{occ}_{\{m,g,s\}}$$

$A_{\text{total},s}$ anchored to the RISC-capacity covariate (`ago_gate`).

4.3 Repression aggregate and observation

$$\rho_{\{g,s\}} = \sum_{\{m \in R(g)\}} w_{\text{eff}}(m,g,s) + \sum_{\{(m,m') \in \text{coop}(g)\}} \psi_{\{g,mm'\}} \cdot \text{occ}_{\{m,g,s\}} \cdot \text{occ}_{\{m',g,s\}}$$

$$Y_{\{s,g\}} = \alpha_g + C_{\{s,\cdot\}} \beta_g - \rho_{\{g,s\}} + \varepsilon_{\{s,g\}}, \quad \varepsilon \sim N(0, \sigma_g^2)$$

β_g (confounder block) **unpenalised**; $\text{coop}(g)$ = arm-pairs with distinct sites 8–40 nt apart.

4.4 Weights, priors, groups (the learned M)

$\theta_{\{g,m\}} = z_{\{g,m\}} \cdot \eta_{\{g,m\}}, \quad \eta_{\{g,m\}} \sim N^+(\mu_{\{g,m\}}, \tau_{\{g,m\}}^2)$ # non-negative magnitude
 $z_{\{g,m\}} \sim \text{Bernoulli}(\pi_{\{g,m\}})$ # spike-and-slab inclusion
 $\eta_{\{g,m\}} = \eta_{\{g,F(m)\}} + \delta_{\{g,m\}}, \quad \delta_{\{g,m\}} \sim N(0, \rho_{F^2})$ # seed-family partial pooling
 $\psi_{\{g,mm'\}} \sim z^{\text{coop}}_{\{g,mm'\}} \cdot N^+(\mu^{\text{coop}}, \tau^2), \quad z^{\text{coop}} \sim \text{Bernoulli}(\pi^{\text{coop}})$ # primed cooperativity,

Priors from evidence fusion (Decision E), **from sequence/curation/external only**:

$\mu_{\{g,m\}} = \log(1 + \sum_{\text{sites}} \text{occ_prior}(K_{\text{site}}))$ # site-level magnitude (context++/K_D)
 $\text{logit } \pi_{\{g,m\}} = w_0 + w_{\text{tier}} \cdot \text{tier} + w_{\text{meth}} \cdot \text{TarBase9_method_conf}(\text{PMID-deduped})$
 $\quad + w_{\text{ago}} \cdot \text{loglp}(\text{AGO-CLIP: ENCORIuPOSTAR3uTarBase-CLIP, PMID-deduped}) + w_{\text{chim}} \cdot \text{loglp}(\text{chimeric: ManakovuTarBase-qCLASH})$
 $\quad + w_{\text{lit}} \cdot \text{loglp}(\text{breast PMIDs}) + w_{\text{seq}} \cdot \text{seq_ind}$ # NB TarBase regulation $\approx$ all-Negative $\rightarrow$ not a direction term
 $\tau_{\{g,m\}} = \tau_0 \cdot \text{decreasing}(\text{evidence_strength})$ # HE tight, orphan loose
 $\text{sign} : \text{from functional assays} + M \geq 0 \text{ prior ; only the } \sim 301 \text{ TarBase Positive edges } \rightarrow \text{activating/ex}$

Source weights w_* **calibrated** against held-out functional repression (not hand-set).

4.5 Estimation

- **Gold:** mean-field VI or Gibbs over $(z, \eta, \delta, \psi, \beta, \sigma)$; report **posterior inclusion probabilities** (discovery score) and posterior widths (family non-identifiability).
- **Surrogate / MVP:** non-negative **adaptive group-elastic-net**: minimise $\sum_{\{g,s\}} (Y - \alpha - C\beta + \rho)^2 + \sum_{\{g,m\}} \lambda_{\{g,m\}} |\theta_{\{g,m\}}| + \text{group penalties}$, with $\lambda_{\{g,m\}} = \lambda/w_{\text{prior}}(g,m)$ (adaptive from μ), seed-family groups, $\theta \geq 0$, β unpenalised. Tune λ, x_0 , pool params by **nested patient-level CV**, never in-sample.

4.6 Where it nests the current heuristic (\$0 sanity anchor)

Freeze $z=1$ on the HE support, $\theta_{\{g,m\}} = \text{edge_w}$ (curated, fixed), linear occupancy ($\text{occ} = a \cdot \theta$), $\phi=1$ (no pool), no cooperative terms, $C=\emptyset$, abundance transform = $\text{expr_mult} \rightarrow$ then $\rho_{\{g,s\}} = \sum_m \text{expr_mult}(m,s) \cdot \text{edge_w}(m,g) = \text{exactly the current pressure}$ (FORMULAS.md §1). So the heuristic is the **infinite-prior-strength, no-confounder, linear-occupancy, hard-HE limit**. Anything learned reads as “the prior plus what the data insists on.”

4.7 Variants kept in play (specified enough to tell apart)

Variant	Change from default	When preferred
Gene-focused adaptive-lasso	one penalised regression per gene, no cross-gene pooling	MVP; deep single-driver dives; direct compare to per-edge coupling
Low-rank RRR/NMF overlay	$M \approx UV^T$, rank r	explain residual covariance; comodule roll-up; borrow max strength
Feature/FM prior generator	$\mu_{\{g,m\}} = \phi(g)^T \theta \psi(m)$	fill μ for non-canonical orphans with no K_D

Variant	Change from default	When preferred
Y = protein / discordance	swap readout; MoPC-style conditioning	function-first genes; translational-class edges; OOD validation
State: separate / differenced	fit M_H, M_T apart, or $\Delta Y = -M \Delta x$	audit; platform-constant subtype-centroid contrast
Subtype environments (ICP)	invariance across PAM50 as a causal filter	causal robustness check; flag subtype-specific edges

5. Evaluation & falsification plan

Pre-named tests, references, success criteria, and abandon conditions. All coupling tests are **partial-Spearman adjusting C**, on **held-out patients** (patient-level CV folds; the prior never touches the tested (X, Y)).

Bar 1 — beat raw abundance (primary, \$0 ref 1). Does the learned **realised aggregate** $\rho_{\{g, s\}}$ anti-correlate with held-out target mRNA better than **raw arm abundance** (and than family-summed abundance)? *Metric:* paired Δ Spearman across folds, partial on C. *Success (pre-named):* median Δ Spearman > 0 with 95% CI excluding 0 across the \$1h driver panel **and** for $\geq$ a pre-registered fraction of program genes. **Abandon-to-abundance** if the learned weights add nothing beyond abundance here — this is the decisive bar.

Bar 2 — match/extend curated fixed-M (\$0 ref 2). Learned-M vs curated-M on the same OOF coupling and on **OOF prediction** of Y (because the coupling-invariance lemma says single-edge coupling can't separate weightings — test at the aggregate/predictive level). *Success:* learned $\geq$ curated (non-inferiority) while **nesting** it, **or** learned discovers ≥ 1 OOD-validated orphan edge curated-M lacks. Beating is *not* required — parsimonious reproduction counts.

Bar 3 — prior does real work (\$0 ref 3). Permute the evidence $\rightarrow (\pi, \mu)$ map (shuffle within degree strata), refit. *Success:* real prior beats shuffled OOF. Guards against “any sparse structure would do.”

Bar 4 — causal, not covariational (the differentiator). miRNA-locus **CN instrument** 2SLS: $x_m = \gamma \cdot \text{CN_locus}(m) + C\delta + u$; $y_g = -M \cdot \hat{x}_m + C\beta + \varepsilon$, on **focal** segments (pleiotropy control). *Report* first-stage F (> 10 = not weak). *Success:* causal M concordant in sign/rough magnitude with the conditioned M for arms with adequate instruments. *Inconclusive* (not failed) where instruments are weak.

Bar 5 — orthogonal layer & independent cohort (anti-circularity crux). Fit on tumour mRNA; score the learned edges, unrefit, on **CPTAC protein** and **protein-vs-mRNA discordance** (MoPC-style partial-corr conditioned on the arm) — the translational residual the mRNA fit never saw; and on an **independent mRNA cohort** (SCAN-B / METABRIC / Buffa). *Success:* learned edges show protein-level repression **and** discordance signal beyond curated-M; independent-cohort coupling replicates in direction. **Manakov chimeric-eCLIP** physical duplex as a binding check on discovered orphans.

Face validity (not a quantitative bar). Recover the \$1h canonical edges: let-7 $\rightarrow$ KRAS/MYC/HMGA2, miR-200 $\rightarrow$ ZEB1/2, miR-29 $\rightarrow$ collagens/DNMT3, miR-21 $\rightarrow$ PTEN/PDCD4, miR-17~92 $\rightarrow$ E2F1/PTEN/BIM, miR-34 $\rightarrow$ CDK4/6/BCL2, miR-221/222 $\rightarrow$ CDKN1B/ESR1. Failure to recover well-instrumented canonical edges is a red flag on the fit.

Circularity guardrail (non-negotiable). π, μ, τ and interaction support from sequence/curation/ external only; expression estimates magnitudes; never test coupling

on the fit's (X,Y); keep learned-M and curated-M side by side and report disagreements as findings.

6. Phased build plan (cheapest, most diagnostic first; each phase gated on a pre-named win)

Companion: `LEARNED_MODEL_BUILD_PLAN.md` maps these phases to the repo — **what to reuse vs rebuild** (the data/infra spine + eval machinery are reused; the model core + evidence ledger are new; the current heuristic is frozen as the baseline; ~100 one-off scripts are archived) and a **target learned/ package architecture** with a low-risk *classify-first, build-alongside, archive-lazily* migration.

Phase 0 — Observation-model audit (cheapest, highest-value). On the existing per-edge coupling, add the **transcription-control block** to C and measure how much canonical edges move (Decision B ablation). *Gate:* the transcription confound is quantified; if adding the proxy flips/kills spurious positive weights on known repressive edges, the block is mandatory downstream. No new estimator yet.

Phase 1 — Gene-focused adaptive-lasso MVP. On a hub panel (PTEN, GATA3, ESR1, ZEB1, IRF1): non- negative adaptive-lasso, $\lambda_{\{g,m\}} = \lambda/w_{\text{prior}}$, seed-family groups, C (with transcription block) unpenalised, occupancy in the unsaturated (linear) approximation. *Gate:* recovers curated drivers **and beats raw-abundance coupling out-of-fold** (Bars 1-2 on the panel).

Phase 2 — Evidence-fusion prior + soft discovery. Build the provenance ledger FIRST. Before any weight-tuning, construct the unified (**m,g,PMID,assay-class**) ledger (miRTarBase + TarBase-v9 + ENCORI + POSTAR3 + Manakov) and **de-duplicate globally by PMID×assay** — physical *and* functional — replacing the current “miRTarBase counts + additive ENCORI” scoring. Then wire biochemical K_D / context++ into (π, μ, τ) and run confirmatory (hard-HE) and discovery (soft) side by side. *Gate:* dedup materially changes $\geq$ some edges’ evidence (sanity: hub CLIP edges collapse toward distinct-PMID counts); Bar 3 (beat shuffled-evidence null); ≥ 1 discovery edge that clears Bar 5 (protein/independent-cohort).

Phase 3 — Program-wise hierarchical model. Full default (\$4) on 3-5 Hallmarks (EMT, P53, G2M, IFN γ , HYPOXIA): family partial pooling, occupancy link + threshold gauge, shared pool. *Gate:* family→gene estimands stable across folds; retire $D(m)$ (pool reproduces promiscuity penalty); OOF coupling $\geq$ curated-M at the aggregate level.

Phase 4 — Cooperativity + ceRNA terms. Add primed product terms (site-coord window) and the shared free-AGO pool depletion. *Gate:* any product/pool term **improves held-out fit** and survives shrinkage; otherwise drop (additive is fine).

Phase 5 — Causal instrument. CN-locus 2SLS on the hub panel and program genes with adequate instruments. *Gate:* first-stage F reported; causal-vs-conditioned concordance for well-instrumented arms (Bar 4).

Phase 6 — State axis. Nested $M_T = M_H + \Delta$ with GTEx prior; test $\Delta=0$. *Gate:* Δ concentrates in APA-shortened / capacity-shifted genes (biology signature) not in GTEx-imputed arms (artifact signature).

Phase 7 — Universe overlay (only if 1-6 pay off). Low-rank RRR/NMF for structural roll-up and comodules; the feature/FM model for non-canonical orphan μ . *Gate:* comodules interpretable; couple OOF.

Each gate mirrors `PRESSURE_FUTURE_OPTIONS.md` discipline: promote a lane only on a pre-named held-out / orthogonal-layer win, never on in-sample fit.

7. Open questions & what would change the answer

1. **Is M actually state-dependent, or only x ?** (Decision H.) The nested model tests $\Delta=0$. If $\Delta \approx 0$, the whole state story collapses to acquired abundance — simpler, and it retires the separate/differenced variants. **What would resolve it:** wiring **APA/PDUI** (Xia tumour, APAAtlas normal) so site loss is an explicit state covariate; if Δ tracks PDUI shortening, M is genuinely state-dependent.
2. **How much does isomiR resolution change the predictor unit?** 5' isomiRs shift the **seed** (positions 2–7) → different family, different targets. This could break some family collinearity (Decision F) and correct the prior's seed. **What would resolve it:** run the arm-level model, then re-run with isomiR- resolved arms on the arms with high 5' heterogeneity; if edges move, isomiR is first-class, not a detail. *Inferred* to matter for a minority of arms **[infer]**.
3. **Does TarBase-v9's direction label change which edges are admissible? — largely answered: no.** I checked the live file: only **301 of 4.72M rows are “Positive”** (the field is an almost-constant “Negative” default, because ~93% of rows are CLIP = binding, not direction). So the activating-edge exclusion is a **tiny** effect, not a major prior channel; the non-negativity constraint is fine for almost all edges. The *open* residual: are those ~301 Positives real activating interactions worth a dedicated class, and does miRTarBase/breast-context add more? **What would resolve it:** map the 301 Positives (and any curated up-regulating edges) onto the Hallmark panel and inspect.
4. **Which observation target wins (Decision B) ripples everywhere.** If protein/discordance turns out to be the only clean readout (transcription confound intractable in mRNA), then: primary Y shifts to CPTAC, n drops, the program-wise pooling (Decision A) becomes *more* important (borrowing strength under small n), and the state axis narrows to states with CPTAC coverage. Conversely if the transcription proxy fully de-confounds mRNA, mRNA stays primary and CPTAC is purely validation.
5. **Is the a-w threshold x_0 estimable, or must it be asserted?** (Decision C.) If OOF tuning finds a stable x_0 , the gauge is data-anchored; if not, it stays a biological assertion (Mullokandov) with a sensitivity band. **What would resolve it:** the CN instrument — a dose perturbation that crosses the threshold would reveal the nonlinearity directly.
6. **Does the shared-pool term truly retire $D(m)$?** (Decision G.) **What would resolve it:** fit with the pool and *without* $D(m)$; check that hub-arm per-target occupancy is depressed to the degree $D(m)$ hand-imposed. If yes, $D(m)$ is gone; if the pool is under-identified, $D(m)$ stays as a fallback.
7. **Family vs arm as the shipped estimand.** If per-arm splits never identify (δ posteriors stay at prior), the honest deliverable is family→gene throughout, and the per-arm “driver nomination” language must be enforced repo-wide. **What would resolve it:** the posterior widths of δ across the driver panel.

Things in the brief I would flag as needing correction or sharpening (§8 item 7): - **Chimeric eCLIP is site-level — but rely on AGO-CLIP as the *primary* CLIP layer** (feedback, 2026-07-04). The brief's §2c lists Manakov under edge-level (“physical proof a pairing occurs”), but chimeric reads carry **duplex coordinates** — so they can **prime cooperative interactions** (Decision G) *and assign the arm*, not just gate inclusion. **However, chimeric is sparse and its reproducibility is contested**, so the design leans on **AGO-CLIP (ENCORI) read depth** — dense, replicated, already in-repo — as the primary physical π signal, with chimeric a *confirmatory* arm-assignment / coordinate layer used only where dense enough, falling back to AGO-CLIP region + seed inference when chimeric is too sparse (Decision E). Their key difference: **AGO-CLIP is miRNA-agnostic** (needs seed inference); **chimeric gives the direct (arm, site) pair**. The design underused AGO-CLIP in its first draft — this corrects it. - **“Site exists” is a weak prior; graded K_D is the strong one.**

The flanking-context ~100-fold affinity range [**assert: McGeary 2019**] means a binary site-type mask under-uses the sequence evidence. The magnitude prior should be the **continuous K_D/occupancy**, not the site-type tier. - **The current heuristic's biggest defect is the missing transcription control**, not the softmax or $D(m)$ per se. Fixing the observation model (Decision B) likely moves more edges than any weight change. - **Composition adjustment is large-effect, not cosmetic — and the metagene under-corrects it** (feedback + repo core_coupling_deconv_retest). Tissue-matched **deconvolution compartment fractions** (the user's CIBERSORTx breast single-cell-atlas run; BayesPrism) are the gold-standard composition covariate; the concrete cost of getting this wrong is the **miR-29b→ECM/collagen** edges collapsing from $-0.3/-0.4$ to ≈ 0 once CAF/stromal fraction is deconvolved. Use compartment fractions for stroma/immune programs and a **malignant-compartment (Cancer-Epithelial) proliferation**, not a bulk metagene (Decision B). - **TarBase v9 is mostly CLIP, not a fresh directional layer** (feedback + live-file check). ~93% of its 4.72M rows are HITS/PAR-CLIP (+ qCLASH chimeric) from **998 PMIDs**, and regulation is **4,724,236 Negative vs 301 Positive** — so it double-counts ENCORI/POSTAR3 CLIP and provides almost no direction. Fold it into the shared PMID-deduped CLIP/chimeric axes; its unique value is site coordinates, confidence, microt_score, and a thin functional layer — **not** direction or extra binding mass. - **The current scores are NOT PMID-deduplicated across databases — and this must be fixed globally** (feedback). evidence_score today = miRTarBase per-class study counts with ENCORI CLIP depth **added on top** (no dedup), and the model's edges table has already collapsed to counts (no PMID column). So the same paper is counted twice both as **CLIP** (ENCORI/POSTAR3/TarBase) and as **functional** (a luciferase/western paper in both miRTarBase and TarBase). Fix = a (m,g,PMID,assay-class) ledger built at edge-build time, deduped by PMID×assay (physical **and** functional), counting distinct PMIDs per class (Decision E). Sequence/K_D priors are PMID-free and stay a separate channel. - **Use several AGO-CLIP sources, not just ENCORI** (feedback). ENCORI AGO-CLIP is sparse (~3% site coverage in the site-ladder); **POSTAR3 carries ~2.36M AGO2 peaks** in-repo and is the dense site-level representative. Keep them non-double-counted (same assay class): ENCORI edge-level, POSTAR3 site-level. - **Metastatic n=7** is too small to model; drop rather than pool. - **The AGO gate is currently non-competitive** (cohort saturating, no pool depletion); ceRNA/promiscuity need the *shared-pool* form (Decision G).

8. References (refreshed July 2026)

Mechanism & magnitude — Guo, Ingolia, Weissman, Bartel (2010) *Nature* “Mammalian miRNAs predominantly act to decrease target mRNA levels”; Eichhorn et al. (2014) *Mol Cell* “mRNA destabilization is the dominant effect... by the time substantial repression ensues” (<https://pmc.ncbi.nlm.nih.gov/articles/PMC4292926/>); Selbach et al. (2008) & Baek et al. (2008) *Nature* pSILAC/proteomics.

Site efficacy & 3'UTR — Agarwal et al. (2015) *eLife* TargetScan7 context++ (<https://www.targetscan.org/>); McGeary, Lin, Shi, Bartel et al. (2019) *Science* “The biochemical basis of microRNA targeting efficacy” (<https://pmc.ncbi.nlm.nih.gov/articles/PMC7051167/>); Soutschek et al. (2022) *Bioinformatics* scanMiR (<https://doi.org/10.1093/bioinformatics/btac110>).

Cooperativity & competition — Saetrom et al. (2007) *NAR* distance constraints (<https://academic.oup.com/nar>); Grimson et al. (2007) *Mol Cell* determinants beyond seed; Broderick et al. (2011) *RNA*; Rinck/Sætrom-line (2013) *RNA* transcriptome enriched for cooperativity-permitting distance (<https://pubmed.ncbi.nlm.nih.gov/23696004/>); Denzler et al. (2014, 2016) *Mol Cell* ceRNA thresholds (<https://pubmed.ncbi.nlm.nih.gov/24793693/>, <https://pmc.ncbi.nlm.nih.gov/articles/PMC5048918/>); Bosson et al. (2014) *Mol Cell* AGO-iCLIP susceptibility (<https://www.sciencedirect.com/science/article/pii/S1546221814000110>); Mullokandov et al. (2012) *Nat Methods* sensor/decoy functional threshold.

Expression modelling — Huang et al. GenMiR++ (2007); Muniategui et al. TaLasso

(<https://www.researchgate.net/publication/51480609>); Zhang et al. (2011) *Bioinformatics* SNMNMF (<https://dx.doi.org/10.1093/bioinformatics/btr206>); PIMiM; Bayesian sparse reduced-rank regression (<https://pmc.ncbi.nlm.nih.gov/articles/PMC5628626/>, <https://pmc.ncbi.nlm.nih.gov/articles/PMC7333421/>); MoPC (2024) mRNA-protein partial correlation conditioned on miRNA (<https://pmc.ncbi.nlm.nih.gov/articles/PMC10956802/>).

Causal / identification — Meng et al. (2019) *BMC Bioinformatics* invariant causal prediction with PAM50 environments (<https://pmc.ncbi.nlm.nih.gov/articles/PMC6419852/>); miRNA Mendelian randomisation (various, 2024–2025, e.g. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC128857/>).

Protein & discordance — CPTAC breast proteogenomics (Mertins et al. 2016; Krug et al. 2020, as reviewed <https://tcr.amegroups.org/article/view/10135/html>); MoPC (above).

State / field effect — epigenetic field cancerisation in breast (medRxiv 19002014); field cancerisation review (<https://pmc.ncbi.nlm.nih.gov/articles/PMC9322418/>); mutational landscape of adjacent normal breast (*Cell Reports Medicine* 2025).

Breast / Hallmark miRNA — subtype-specific miRNA & AGO2-PAR-CLIP subtype target regulation (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4053773/>); subtype miRNA networks review (<https://www.mdpi.com/2227-9059/10/3/651>).

CLIP / chimeric — Helwak et al. (2013) *Cell* CLASH (<https://pmc.ncbi.nlm.nih.gov/articles/PMC3650559/>); chimeric eCLIP / miR-eCLIP (bioRxiv 2022.02.13.480296; *eclipsebio*); mirTarCLASH (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11971479/>).

Evaluation & overfitting — Pinzón et al. (2017) *Genome Res* “microRNA target prediction programs predict many false positives” (<https://pmc.ncbi.nlm.nih.gov/articles/PMC5287229/>); Kern et al. (2021) NAR validation of miRNA target pathways (<https://academic.oup.com/nar/article/49/1/127/>).

Prepared as the §9 deliverable for LEARNED_MODEL_DISCUSSION_PROMPT.md. Reasoning-first: every decision carries its mechanism, its grounding in biology/data/identifiability, the alternatives kept in play with when each wins, and its failure modes. The recommended default nests the current heuristic and is falsifiable against raw abundance, curated-M, and shuffled-evidence nulls.